

Answer **all** questions in the spaces provided

- 1** An aeroplane is at rest on a runway. It accelerates in a straight line along the runway and takes off after 55.0 s.

Fig. 1.1 shows the variation with time of the resultant force acting on the aeroplane while it is in contact with the runway.

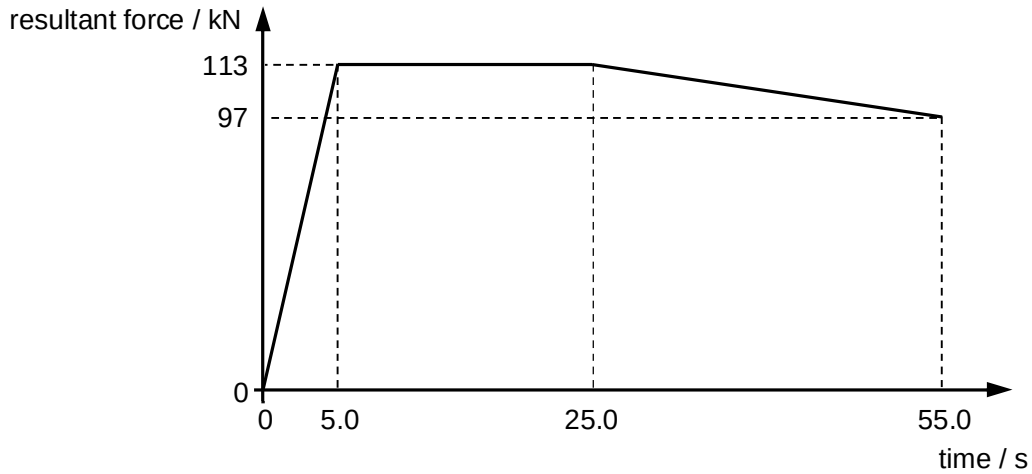


Fig. 1.1

- (a)** State two reasons why the resultant force on the aeroplane is not constant as it travels along the runway.

1.

.....

2.

..... [2]

- (b)** Calculate the momentum of the aeroplane at take-off.

momentum = N s [3]

(c) The total mass of the aeroplane is 7.45×10^4 kg .

Calculate the velocity $v_{\max}$ of the aeroplane at take-off.

$v_{\max} = \dots\dots\dots \text{ m s}^{-1}$ [1]

(d) On Fig. 1.2, sketch a graph to show the variation with time of the velocity of the aeroplane as it travels along the runway. (Numerical values for the velocity are not required.)

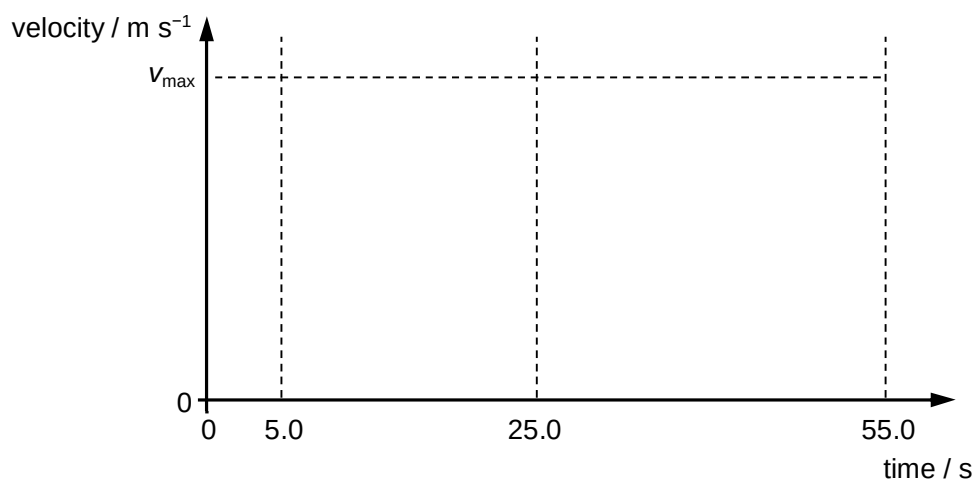


Fig. 1.2

[2]

(e) Estimate the distance the aeroplane travels along the runway.

distance = $\dots\dots\dots$ m [2]